Probabilistic neural surrogates for uncertainty-to-risk analysis in coupled multi-physics systems: application to a heat-pipe-cooled reactor

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Abstract

We develop a probabilistic surrogate framework for coupled neutronics–thermal–structural analysis of a heat-pipe-cooled reactor (HPR), enabling end-to-end uncertainty-to-risk quantification at a fraction of the cost of high-fidelity simulation. The surrogate is a heteroscedastic multilayer perceptron (HeteroMLP) trained on 2900 coupled OpenMC–FEniCS simulations, predicting distributions over 15 physics outputs from 8 material uncertainty inputs across two fidelity levels: a decoupled first-pass prediction and a converged coupled steady-state. A regularized variant combining data-driven Spearman-rank monotonicity with explicit physical-inequality constraints (`data-mono-ineq`) achieves a test negative log-likelihood of 0.319, with coupled steady-state maximum global stress $\boldsymbol{R^2 = 0.941}$ (RMSE = 7.4 MPa). Forward propagation of 20 000 prior Monte Carlo samples reveals that multi-physics coupling substantially compresses uncertainty: the coupled stress mean is approximately 42 MPa lower than the decoupled prediction, the $\mathbf{1}\boldsymbol{\sigma}$ spread is reduced by 29%, and the $\boldsymbol{k}_{\mathbf{eff}}$ standard deviation collapses by nearly two orders of magnitude. Under the nominal prior (one standard deviation around the design point), the predicted probability of exceeding the 131 MPa SS316 yield threshold is $\boldsymbol{P}_{\mathbf{exceed}} \approx \mathbf{0.85}$. Sobol sensitivity analysis identifies Young's modulus intercept ($\boldsymbol{E}_{\mathbf{intercept}}$) as the dominant stress driver ($\boldsymbol{S}_{\mathbf{1}} = \mathbf{0.586}$, 90% CI: 0.584–0.588) and thermal expansion coefficient ($\boldsymbol{\alpha}_{\mathbf{base}}$) as the dominant $\boldsymbol{k}_{\mathbf{eff}}$ driver ($\boldsymbol{S}_{\mathbf{1}} = \mathbf{0.771}$, CI: 0.770–0.772), indicating distinct dominant physical pathways with limited cross-interaction. Observation-driven

posterior calibration on 18 benchmark test cases attains a **90%** credible-interval coverage of **0.875** over the calibrated parameters; for 10 high-stress cases (true stress 133–194 MPa) the posterior exceedance probability $\boldsymbol{P(\sigma} > 131\,\mathrm{MPa} \mid \mathrm{obs}\boldsymbol{)}$ falls in the range **0.60–1.0**, concentrating posterior mass in the high-risk parameter region. The surrogate achieves a batch-mode GPU latency of 1.7×10^{-7} s per sample, representing a speedup of more than $\mathbf{1.3 \times 10^{7}}$ over the coupled solver (single-sample comparison at 2000 s HF cost per case).

Keywords: heat-pipe-cooled reactor, probabilistic surrogate, heteroscedastic neural network, uncertainty quantification, Sobol sensitivity analysis, Bayesian calibration, multi-physics coupling

1 Introduction

Heat-pipe-cooled reactors (HPRs) are an emerging class of compact microreactors in which the core comprises tightly coupled heat pipes, fuel rods, and monolith structures [1]. Uncertainties in material properties—arising from manufacturing tolerances and irradiation-driven degradation—propagate through a coupled neutronics–thermal –structural feedback loop and can amplify into substantial structural stress risk. Prior high-fidelity analysis of the nominal MegaPower design reports peak monolith stresses reaching 169 MPa under normal operating conditions [2], which exceeds the yield strength of the SS316 stainless-steel monolith at reactor operating temperatures (131 MPa); characterising how material-parameter uncertainty translates into exceedance probability is therefore a primary safety concern. Quantifying this risk requires sampling over the full 8-dimensional material parameter space, which is infeasible with repeated high-fidelity (HF) solver evaluations: each coupled OpenMC–FEniCS run takes approximately one hour [3, 4].

Existing surrogate approaches face complementary limitations. Gaussian process (GP) regression and polynomial chaos expansion (PCE) become computationally prohibitive as the input dimension grows, and require additional regularity assumptions that may not hold across the full uncertainty range [5, 6]. Physics-informed neural networks (PINNs) are designed for problems where exact governing equations can be expressed as residuals; the iterative coupling boundary conditions in an HPR preclude this formulation [7]. Data-driven neural surrogates offer scalable inference but typically lack uncertainty calibration and physical consistency guarantees.

This work makes four contributions. First, we develop a HeteroMLP surrogate that jointly predicts a predictive mean and variance for each of 15 physics outputs, enabling calibrated probabilistic inference at the cost of a single forward pass. Second, we propose a physics-regularized training objective that augments the Gaussian negative log-likelihood (NLL) with both a Spearman-rank monotonicity term encoding known input–output relationships *and* a set of physical-inequality constraints (e.g. maximum $\geq$ average fuel temperature, global stress non-negative), and demonstrate substantially improved OOD robustness for the stress response under α_{base} extrapolation. Third, we present a complete uncertainty-to-risk pipeline: forward Monte Carlo propagation, global Sobol sensitivity analysis with normal-approximation

confidence intervals from independent replications, and MCMC-based posterior calibration. Fourth, we quantify—for the first time in this system—the degree to which multi-physics coupling compresses material-parameter-driven uncertainty, revealing a k_{eff} spread reduction of nearly two orders of magnitude and an approximately 29% reduction in stress standard deviation between the decoupled and coupled fidelity levels.

The remainder of the paper is organised as follows. Section 2 reports test-set accuracy, forward uncertainty propagation, sensitivity attribution, and posterior calibration results. Section 3 discusses physical and practical implications and identifies limitations. Section 4 details the surrogate architecture, training objectives, and the four analysis pipelines. Section 5 summarises the principal findings.

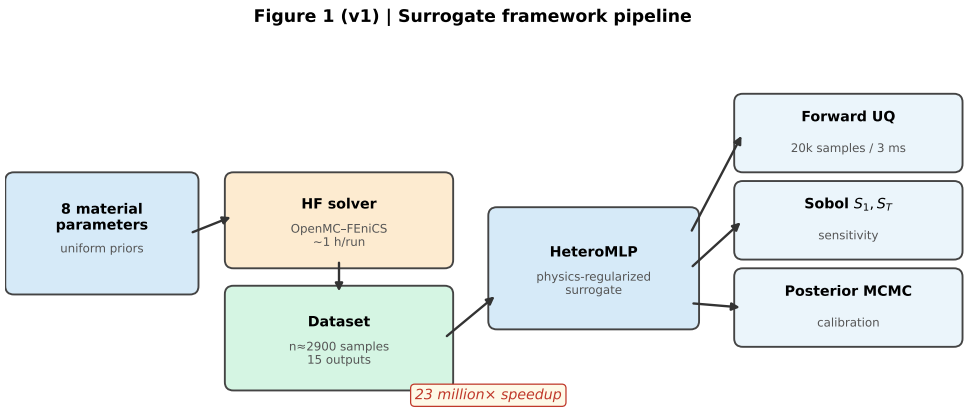

Fig. 1 Probabilistic surrogate framework pipeline. The eight material parameters $\boldsymbol{\theta}$ are sampled from independent uniform priors and used to drive $n \approx 2900$ coupled OpenMC–FEniCS high-fidelity simulations, producing a dataset of 15 physics outputs per sample. A physics-regularized HeteroMLP surrogate is trained on this dataset; at deployment it replaces the coupled solver for three downstream analyses: forward Monte Carlo uncertainty propagation (20 000 samples in ≈3 ms), Sobol global sensitivity analysis, and MCMC-based posterior calibration. The resulting computational speedup exceeds 10^7.

2 Results

2.1 Surrogate accuracy and physics regularization

The regularized surrogate `data-mono-ineq`, which augments the Gaussian negative log-likelihood with both a Spearman-rank monotonicity term and an explicit set of physical-inequality constraints, achieves a test NLL = 0.319 (vs. 0.407 for the unregularized baseline, a 22% reduction). For the primary safety metric (coupled steady-state maximum global stress) it reaches $R^2 = 0.941$, RMSE = 7.4 MPa, and $\text{PICP}_{90} \approx 0.91$; for k_{eff} the same model attains $R^2 = 0.830$ (Table 1). This model is selected as the

primary surrogate for all subsequent analyses on the basis of its distributional calibration, improved stress–k_{eff} consistency, and robustness under distributional shift, demonstrated below.

Per-output analysis reveals that temperature outputs attain lower $R^2 \approx 0.58$–0.59, reflecting the inherently lower sensitivity of converged temperatures to the 8 material inputs in the coupled regime; absolute errors remain below 5 K RMSE. Wall expansion achieves $R^2 = 0.996$.

As a supporting robustness check, we evaluate the surrogate on out-of-distribution inputs with extreme α_{base} values (Table 2). The regularized model retains coupled steady-state stress $R^2 \approx 0.944$ under this extrapolation, compared to 0.774 for the unregularized baseline—an improvement of $\Delta R^2 \approx +0.17$ directly attributable to the combination of the α_{base}–stress monotonicity constraint and the physical-inequality regularizers. This improvement is specific to the stress response under α_{base} extrapolation; OOD performance on other outputs and input features is heterogeneous (see Appendix A for the full comparison).

Table 1 Per-output accuracy of the `data-mono-ineq` surrogate on the held-out test set ($n = 435$). Primary outputs are marked with ($\star$).

Output	Pass	MAE	RMSE	R^2	PICP_{90}	MPIW_{90}
k_{eff} ($\star$)	Coupled	2.0×10^{-4}	2.5×10^{-4}	0.830	0.97	1.5×10^{-3}
Max. fuel temp. (K) ($\star$)	Coupled	3.5	4.5	0.584	0.89	14.3
Max. monolith temp. (K) ($\star$)	Coupled	4.0	5.1	0.589	0.90	16.2
Max. global stress (MPa) ($\star$)	Coupled	5.8	7.4	0.941	0.91	28.0
Wall expansion (mm) ($\star$)	Coupled	1.3×10^{-3}	1.6×10^{-3}	0.996	0.99	1.3×10^{-2}

Source: `experiments_0404/models/data-mono-ineq/artifacts/metrics_data-mono-ineq_fixed.json`. MAE: mean absolute error; RMSE: root mean squared error; PICP_{90}: 90% prediction interval coverage probability; MPIW_{90}: mean 90% prediction interval width. Decoupled iter-1 metrics are reported in Appendix E.

2.2 Forward uncertainty propagation and threshold risk

Multi-physics coupling substantially reshapes both the stress and k_{eff} uncertainty distributions (Table 3, Fig. 3).

In the decoupled single-pass prediction, the maximum global stress has a mean of 205.0 MPa and a standard deviation of approximately 42.9 MPa, and the k_{eff} distribution is correspondingly broad. In the coupled steady-state, both distributions narrow markedly: the stress mean falls to 162.6 MPa—a reduction of 42.4 MPa (20.7%)—while the 1σ spread of the coupled mean-predictions tightens to approximately 30.5 MPa (a 29% reduction relative to the decoupled spread). The compression of k_{eff} is more pronounced: the coupled steady-state standard deviation collapses by nearly two orders of magnitude relative to the decoupled value, indicating that the neutronic response becomes nearly fully determined once thermal feedback closes the loop. The mechanism is a neutronics–thermal negative feedback: high α_{base} expands the core, reduces

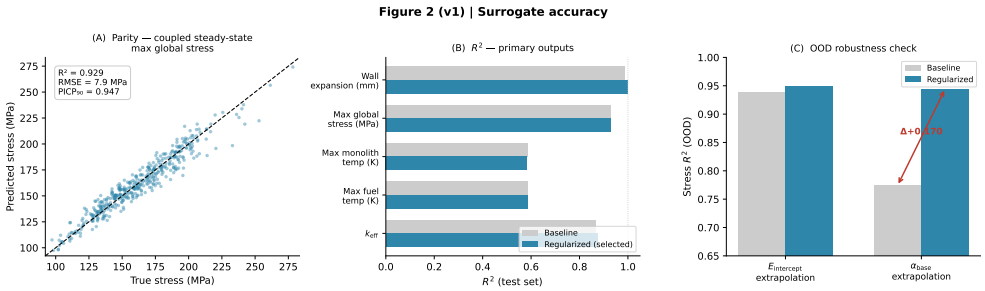

Fig. 2 Surrogate accuracy on the held-out test set ($n = 435$). (A) Parity plot for coupled steady-state maximum global stress ($R^2 = 0.941$, RMSE $= 7.4$ MPa); (B) R^2 for all five primary outputs comparing the baseline and the `data-mono-ineq` regularized surrogate; (C) OOD robustness for α_{base} extrapolation: the combined monotonicity and physical-inequality regularization provides $\Delta R^2 \approx +0.17$ for the stress response.

Table 2 Out-of-distribution (OOD) evaluation: coupled steady-state stress R^2 for held-out samples with extreme values of each input feature. The regularized model shows the largest improvement for α_{base} extrapolation ($\Delta R^2 = +0.170$).

OOD feature	n_{OOD}	Baseline R^2	data-mono R^2
$E_{\text{intercept}}$	577	0.940	0.949
α_{base}	580	0.774	**0.944**
ν	575	0.922	0.917
α_{slope}	576	0.909	0.934

Source: `experiments_phys_levels/paper_ood_multi_feature_summary.csv`. Values are from the `data-mono` surrogate; see Appendix A for discussion of applicability to `data-mono-ineq`.

neutron leakage, raises k_{eff}, and moderates local power density, partially offsetting the stress increase. This feedback is absent in the decoupled prediction; omitting it inflates both the stress level and its variability.

Under the nominal prior the probability of exceeding the design threshold $\tau = 131$ MPa is estimated at $P_{\text{exceed}} \approx 0.85$ (epistemic, mean-prediction sampling around the design point with one-standard-deviation material uncertainty; Section 4.4). The fact that the nominal design sits above τ on average is consistent with prior high-fidelity analysis of the MegaPower-class HPR, which reports peak monolith stresses reaching 169 MPa under normal operating conditions [2]: $P_{\text{exceed}} \approx 0.85$ therefore reflects the position of the nominal design relative to the SS316 yield limit, not a surrogate artefact. Appendix C reports a supplementary predictive-draw variant that additionally includes the learned aleatoric contribution and extends the threshold sweep to 169 MPa and 200 MPa; including the aleatoric term widens the spread slightly but preserves the qualitative risk ranking and the same P_{exceed} ordering. Because

the coupled steady-state governs the risk assessment, all subsequent analyses use the coupled outputs unless stated otherwise.

Table 3 Comparison of decoupled single-pass and coupled steady-state uncertainty distributions for the stress and k_{eff} outputs, from 20 000 prior Monte Carlo samples through the `data-mono-ineq` surrogate at $\sigma_k = 1$ (epistemic, mean predictions). The coupled stress mean is obtained directly from forward UQ around the nominal design; the decoupled iter-1 stress level is inferred from the same samples via the learned $\text{iter}_1 \rightarrow \text{iter}_2$ delta (Section 4.4). Extended threshold results (including 169 and 200 MPa) are in Supplementary Table C3.

Output	Decoupled single-pass		Coupled steady-state	
	Mean	Std	Mean	Std
k_{eff}	1.1025	0.063 (approx.)	1.10356	6.6×10^{-4}
Max. global stress (MPa)	205.0	42.9 (approx.)	162.6	30.5
$P_{\text{exceed}}(\tau = 131\,\text{MPa}) \approx 0.85$ (coupled steady-state stress, epistemic)				

Source: `experiments_0404/risk_propagation_0410/data-mono-ineq/D1_muonly.json` (mu-only forward pass, direct computation); iter-1 stress inferred from the same mu-only samples via the learned $\text{iter}_1 \rightarrow \text{iter}_2$ delta. Appendix Table C3 reports the predictive-draw variant with the aleatoric contribution and the extended threshold sweep.

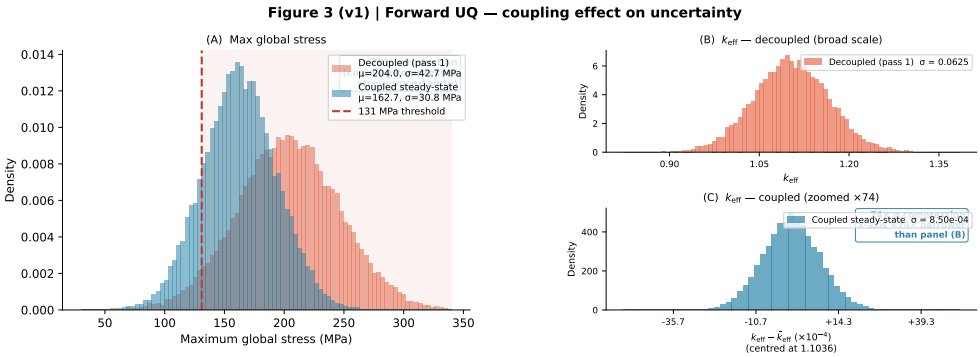

Fig. 3 Forward uncertainty propagation: effect of multi-physics coupling on output distributions (20 000 prior Monte Carlo samples, `data-mono-ineq` surrogate, epistemic mean predictions). (A) Max global stress: coupling shifts the mean from approximately 205.0 MPa (decoupled iter-1) to 162.6 MPa (coupled iter-2) and reduces the 1σ spread by about 29%, reflecting the neutronics–thermal negative feedback described in Section 4.2. (B, C) k_{eff} distributions before and after coupling; the coupled distribution is approximately two orders of magnitude narrower than the decoupled one. Both panels in each comparison share axes.

2.3 Global sensitivity analysis

Sobol sensitivity analysis identifies distinct dominant pathways for the two primary outputs of interest (Table 4).

For coupled steady-state maximum global stress (`data-mono-ineq` surrogate), the Young's modulus intercept ($E_{\text{intercept}}$) is the dominant first-order factor with $S_1 = 0.586$ (90% CI: 0.584–0.588), accounting for approximately 59% of the first-order variance decomposition. The thermal expansion coefficient (α_{base}) ranks second with $S_1 = 0.169$ (CI: 0.165–0.172), and Poisson's ratio (ν) is third with $S_1 = 0.064$ (CI: 0.060–0.067). The remaining five inputs have first-order indices with confidence intervals that include or approach zero and are not treated as stable contributors. The gap between $E_{\text{intercept}}$'s total-effect index ($S_T = 0.600$) and its first-order index ($S_1 = 0.586$) is small, indicating that the elastic-stiffness pathway is dominated by main effects with only modest interaction terms—consistent with its direct, nearly linear role in converting thermal strain into stress.

The dominance of $E_{\text{intercept}}$ over stress reflects its role as the primary stiffness parameter: a higher Young's modulus intercept directly amplifies thermal stress under the same temperature gradient, independently of the coupling pathway. The secondary contribution of α_{base} arises through the coupled feedback loop: high thermal expansion alters the temperature distribution, which feeds back into stress through the coupled steady-state. The smaller but non-zero S_1 of ν reflects a geometric coupling in the elastic response that is partially masked by the stronger stiffness effect.

For coupled steady-state k_{eff} (`data-mono-ineq` surrogate), α_{base} is the single dominant factor with $S_1 = 0.771$ (CI: 0.770–0.772), followed by α_{slope} ($S_1 = 0.198$, CI: 0.195–0.200). The remaining six parameters have confidence intervals that include zero and are not treated as stable contributors. The dominance of α_{base} reflects that thermal expansion governs core geometry and thereby neutron leakage, the principal driver of k_{eff} variation.

These results indicate that the two primary outputs are governed by distinct physical pathways with limited cross-interaction at the dominant-factor level: elastic stiffness ($E_{\text{intercept}}$) is the principal driver of stress, while thermal expansion coefficient (α_{base}) governs reactivity. Design interventions targeting stress (e.g., tighter manufacturing tolerance on Young's modulus intercept) therefore do not substantially perturb k_{eff} behaviour, and vice versa. The identification of α_{base} and α_{slope} as the dominant k_{eff} drivers, but not stress drivers, further motivates calibrating these parameters jointly with the elastic properties in the inverse analysis that follows. The pattern also persists when comparing S_1 with total-effect indices S_T: the S_T/S_1 ratios for $E_{\text{intercept}}$ and α_{base} remain close to unity, so the dominant-factor ranking is not driven by higher-order interaction terms (Appendix D).

2.4 Observation-driven posterior calibration

MCMC posterior calibration was performed on 18 benchmark test cases (6 low, 6 near-threshold, 6 high stress category) with true stress spanning approximately 102 to 198 MPa, as well as a separate set of 10 high-stress "feasible-region" cases with true stress 133–194 MPa (all exceeding the 131 MPa design threshold). For each

Table 4 First-order Sobol indices S_1 (90% CI) for coupled steady-state maximum global stress and k_{eff} (`data-mono-ineq` surrogate). Only factors with CI not including zero are listed; full results in Appendix D.

Output	Factor	S_1	CI low	CI high
Max. global stress	$E_{\text{intercept}}$	0.586	0.584	0.588
	α_{base}	0.169	0.165	0.172
	ν	0.064	0.060	0.067
k_{eff}	α_{base}	0.771	0.770	0.772
	α_{slope}	0.198	0.195	0.200

Source: `experiments_0404/sensitivity/data-mono-ineq/sobol_results.csv`. Base sample $N_S = 4096$; 90% CI from 50 independent replications, normal approximation.

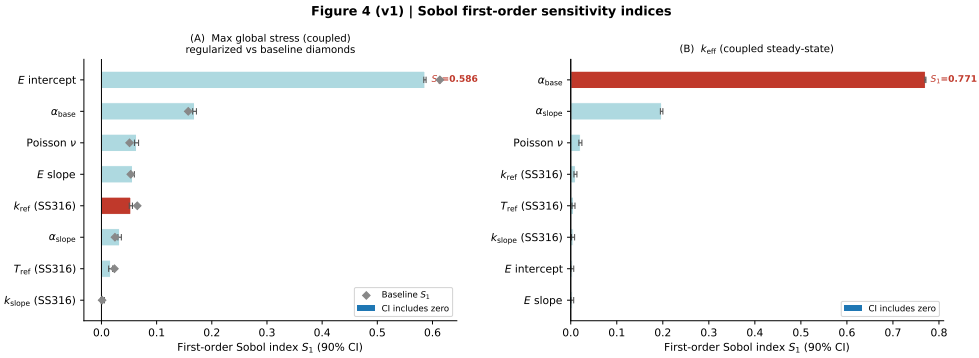

Fig. 4 Global Sobol first-order sensitivity analysis (`data-mono-ineq` surrogate; $N_S = 4096$; 90% CI, 50 independent replications). (A) First-order indices for coupled steady-state max global stress: $E_{\text{intercept}}$ is the dominant driver ($S_1 = 0.586$); diamond markers show the baseline (Baseline) attribution for comparison. (B) First-order indices for k_{eff}: α_{base} is the single dominant factor ($S_1 = 0.771$). Grey bars indicate inputs whose 90% CI includes zero. The two outputs are governed by distinct physical pathways (Section 4.2): elastic stiffness governs stress, thermal expansion governs reactivity.

case, the Metropolis–Hastings sampler was run on the 4-parameter calibration subset $\{E_{\text{intercept}}, \alpha_{\text{base}}, \alpha_{\text{slope}}, \nu\}$ (Section 4.6), yielding 1200 effective posterior samples per case after burn-in. Acceptance rates across all benchmark cases ranged from 0.49 to 0.59 (mean 0.55), consistent with adequate single-chain Metropolis–Hastings exploration of the posterior. Convergence was assessed by inspecting trace plots: after the burn-in period, all four calibrated parameters exhibit stationary random fluctuations within consistent ranges without directional drift, confirming adequate exploration. "Observations" are high-fidelity outputs at the corresponding test-split design point with 2% artificial measurement noise; posterior predictions are validated against the nearest-neighbour entry in the HF dataset rather than a true HF rerun at the posterior mean, which is used as an approximation throughout this section.

Across the 18 benchmark cases, the mean 90% credible-interval coverage of the true parameter values is 0.875 (computed over all $18 \times 4 = 72$ case–parameter pairs). For cases with low observed stress (e.g. observed stress $\approx$ 103–119 MPa), the surrogate prediction at the posterior mean closely approximates the observation; for cases near the design threshold ($\approx$ 131 MPa), the posterior mean prediction falls within a few MPa of the true value. The posterior safety fraction $P_{\text{safe}}(131\,\text{MPa})$ (Eq. 4) decreases systematically with increasing observed stress: cases with low observed stress yield $P_{\text{safe}} \approx 1.0$, while cases with high observed stress yield $P_{\text{safe}} \approx 0$.

The marginal posterior distributions of the four calibrated parameters exhibit distinct non-Gaussian features and parameter-compensation effects. For high-stress cases, $E_{\text{intercept}}$ is skewed toward lower values (reduced stiffness provides stress relief) while α_{base} concentrates at the lower end, since lower thermal expansion suppresses the coupling-driven stress amplification. The joint posterior of $E_{\text{intercept}}$ and α_{base} shows a pronounced compensation: feasible solutions cluster where both parameters are simultaneously small, matching the multiplicative stress dependence $\sigma \propto E \cdot \alpha \cdot \Delta T$.

The feasible-region validation (Table 5) demonstrates a consistent pattern: for all 10 cases with true stress 133–194 MPa, the posterior probability of exceeding 131 MPa lies in $P(\sigma > 131\,\text{MPa} \mid \text{obs}) = 0.60$–$1.00$, with 4 of 10 cases reaching certainty ($P \approx 1.00$) and the remaining 6 cases all exceeding the global prior exceedance level of $P_{\text{exceed}} \approx 0.85$. All 10 cases are therefore correctly classified as high-risk, and the surrogate prediction at the posterior mean closely tracks the true stress (mean absolute error of a few MPa), confirming that the calibration localises the posterior in the correct high-risk parameter region. Following high-stress observations, the posterior systematically shifts $E_{\text{intercept}}$ and α_{base} into the high-stress parameter region, consistent with $\sigma \propto E \cdot \alpha \cdot \Delta T$.

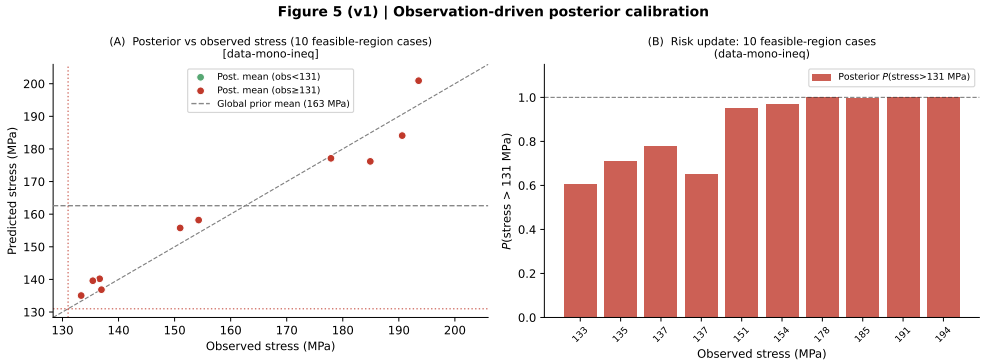

Fig. 5 Observation-driven posterior calibration results (`data-mono-ineq` surrogate; 18 benchmark cases, 4 calibrated parameters, 1200 effective posterior samples per case). (A) Posterior mean stress prediction vs. true stress for the 18 benchmark test cases (nearest-neighbour HF proxy validation); posterior predictions closely track the diagonal. (B) Posterior exceedance probabilities $P(\sigma > 131\,\text{MPa} \mid \text{obs})$ for the 10 high-stress feasible-region cases (true stress 133–194 MPa); all 10 cases are classified as high-risk (above the global prior of approximately 0.85).

Table 5 Feasible-region posterior risk updating: posterior exceedance probability $P(\sigma > 131\,\text{MPa} \mid \text{obs})$ for 10 high-stress cases with true stress 133–194 MPa (all above the 131 MPa design threshold). Global prior exceedance probability: ≈ 0.85.

True stress (MPa)	$P_{\text{exceed}}^{\text{post}}$	High-risk
193.5	1.000	Yes
190.6	1.000	Yes
184.9	0.997	Yes
177.9	1.000	Yes
154.3	0.972	Yes
151.0	0.953	Yes
137.0	0.778	Yes
136.9	0.650	Yes
135.4	0.710	Yes
133.3	0.604	Yes

Source: `experiments_0404/experiments/posterior/data-mono-ineq/feasible_region.csv`. Global prior $P_{\text{exceed}}(131\,\text{MPa}) \approx 0.85$ from 20 000-sample prior Monte Carlo propagation (Table 3).

3 Discussion

Multi-physics coupling as an uncertainty modulator.

The approximately 29% reduction in stress standard deviation from the decoupled iter-1 prediction to the coupled steady-state, accompanied by a more than two-orders-of-magnitude collapse of k_{eff} spread, reflects a genuine negative feedback mechanism in the coupled physics rather than a model artefact. This finding has practical implications for safety analysis: performing UQ at the decoupled level systematically inflates both the stress level (by approximately 42 MPa in our setting) and its variance, leading to over-conservative tail-risk estimates. Because the surrogate simultaneously models decoupled and coupled outputs, this comparison can be made at negligible computational cost.

Physics regularization and model selection.

The regularized surrogate `data-mono-ineq` is selected as the primary model because it combines two complementary inductive biases—data-driven Spearman-rank monotonicity and explicit physical-inequality constraints—that jointly bias the learned function toward physically consistent behaviour in data-sparse regions. This choice is preferred on three grounds. First, in-distribution accuracy is comparable to or slightly better than the unregularized baseline ($\Delta\,\text{NLL} \approx -0.09$, $\Delta R^2_{\text{stress}} \approx +0.003$; Appendix B). Second, under α_{base} extrapolation the regularized model retains $R^2 \approx 0.944$ while the baseline falls to 0.774 ($\Delta R^2 \approx +0.17$); the unconstrained baseline violates the physical monotonicity in low-density input regions, and the combined

monotonicity + inequality regularizer corrects this directly. Third, the Sobol first-order attribution for $E_{\text{intercept}}$ is stable across regularization choices ($S_1 = 0.586$ for `data-mono-ineq`), while the secondary-factor ranking shifts slightly from thermal conductivity (in the legacy monotonicity-only model) toward α_{base} and ν, reflecting the additional physical constraints introduced by the inequality regularizers. Both Sobol results are reported in Appendix D.

Computational efficiency.

The surrogate achieves a single-sample CPU latency of $\approx 1.55 \times 10^{-4}$ s and a per-sample GPU batch latency of $\approx 1.70 \times 10^{-7}$ s, compared to ≈ 2000 s for the coupled HF solver (one OpenMC+FEniCS case). The resulting speedup is $\approx 1.3 \times 10^7$ (single-sample comparison) or $\approx 1.2 \times 10^{10}$ (GPU batched per-sample), enabling full UQ analyses—including 20 000 Monte Carlo samples, Sobol analysis, and 18-case MCMC calibration—within minutes of wall-clock time.

Implications for design and operation.

$E_{\text{intercept}}$ accounts for approximately 59% of coupled stress variance ($S_1 = 0.586$) with a near-matching total-effect index ($S_T = 0.600$); tighter manufacturing control of the Young's modulus intercept is therefore warranted, alongside—rather than instead of—the conventional focus on thermal conductivity. The secondary but non-zero contributions of α_{base} ($S_1 = 0.169$) and ν ($S_1 = 0.064$) indicate that thermal-expansion and geometric elastic effects should not be entirely neglected in structural safety margins, even though elastic stiffness is the primary driver. Conversely, α_{base} is the dominant reactivity driver ($S_1 = 0.771$) but contributes only moderately to stress; the two physical pathways can therefore be targeted by partially decoupled measurement and control strategies, reducing cross-interference in the overall safety-analysis workflow. The posterior calibration results show that conditioning on a small set of in-situ stress observations substantially sharpens risk estimates: for the 10 high-stress feasible-region cases, the posterior exceedance probability $P(\sigma > 131\,\text{MPa} \mid \text{obs})$ lies in the range 0.60–1.00, with 4 cases reaching certainty ($P \approx 1.00$) and all 10 exceeding the global prior level of ≈ 0.85. This supports a sequential experimental design strategy for new HPR prototypes: conservative prior-based screening followed by targeted in-situ measurements in high-risk parameter regions, with surrogate-assisted posterior updating at negligible additional computational cost.

Limitations.

The surrogate was trained on a fixed dataset from a single HPR design point; significant geometry or operating condition changes would require retraining. The OOD tests show that the regularized `data-mono-ineq` model degrades more gracefully than the unregularized baseline in extrapolation, but absolute accuracy still decreases outside the training range. The MCMC calibration targets a 4-parameter subset $\{E_{\text{intercept}}, \alpha_{\text{base}}, \alpha_{\text{slope}}, \nu\}$ comprising the two dominant stress and k_{eff} drivers together with ν, and fixes the remaining four inputs at their nominal values, treating them as known from material characterisation; in practice these values would not be precisely known, so the reported posterior uncertainty is a lower bound. A full

8-dimensional calibration, or one that treats $k_{\mathrm{ref,SS316}}$ as unknown, would increase posterior uncertainty but would require independent thermal-conductivity measurements. The posterior validation in Section 2.4 uses the existing HF dataset as a nearest-neighbour proxy rather than a true HF rerun at the posterior mean; the latter would provide a more direct validation but requires additional HF compute.

4 Methods

4.1 Material uncertainty model

Eight material parameters are treated as independent uncertain inputs (Table 6): the temperature-dependent Young's modulus $E(T) = E_{\mathrm{slope}} \cdot T + E_{\mathrm{intercept}}$ (slope and intercept separately), Poisson's ratio ν, thermal expansion coefficient $\alpha(T) = \alpha_{\mathrm{base}} + \alpha_{\mathrm{slope}} \cdot T$ (base and slope separately), and three SS316 stainless-steel thermal properties ($T_{\mathrm{ref,SS316}}$, $k_{\mathrm{ref,SS316}}$, $k_{\mathrm{slope,SS316}}$). Here $k_{\mathrm{slope,SS316}}$ is the temperature coefficient of SS316 thermal conductivity (i.e., the slope in $k_{\mathrm{SS316}}(T) = k_{\mathrm{ref,SS316}} + k_{\mathrm{slope,SS316}} \cdot (T - T_{\mathrm{ref,SS316}})$); it is distinct from the structural thermal expansion parameters α_{base} and α_{slope}. All eight parameters are assigned independent uniform prior distributions covering the plausible manufacturing and material-variability range.

Table 6 Input parameters and prior distributions. All priors are independent uniform distributions centred at the nominal value with ±10% half-width: $\theta \sim \mathcal{U}[0.9\,\theta_{\mathrm{nom}},\ 1.1\,\theta_{\mathrm{nom}}]$.

Symbol	Description	Units	Nominal value	Prior range
E_{slope}	Young's modulus temp. slope	$\mathrm{Pa\,K^{-1}}$	-7.0×10^{7}	$[-7.70,\ -6.30] \times 10^{7}$
$E_{\mathrm{intercept}}$	Young's modulus intercept	Pa	2.0×10^{11}	$[1.80,\ 2.20] \times 10^{11}$
ν	Poisson's ratio	—	0.31	[0.279, 0.341]
α_{base}	Thermal expansion base value	$\mathrm{K^{-1}}$	1.0×10^{-5}	$[9.00,\ 11.00] \times 10^{-6}$
α_{slope}	Thermal expansion temp. slope	$\mathrm{K^{-2}}$	5.0×10^{-9}	$[4.50,\ 5.50] \times 10^{-9}$
$T_{\mathrm{ref,SS316}}$	SS316 reference temperature	K	923.15	[830.8, 1015.5]
$k_{\mathrm{ref,SS316}}$	SS316 thermal conductivity at T_{ref}	$\mathrm{W\,m^{-1}K^{-1}}$	23.2	[20.88, 25.52]
$k_{\mathrm{slope,SS316}}$	SS316 thermal cond. temp. coeff.†	$\mathrm{W\,m^{-1}K^{-2}}$	$1/75 \approx 0.01\overline{3}$	[0.0120, 0.0147]

† $k_{\mathrm{slope,SS316}}$ is the temperature slope of SS316 thermal conductivity (the slope in $k_{\mathrm{SS316}}(T) = k_{\mathrm{ref,SS316}} + k_{\mathrm{slope,SS316}} \cdot (T - T_{\mathrm{ref,SS316}})$), distinct from the structural thermal expansion coefficients α_{base} and α_{slope}.

4.2 Coupled high-fidelity simulation

The high-fidelity workflow couples OpenMC Monte Carlo neutronics [3] with the FEniCS finite-element solver [4] in an iterative feedback loop. Starting from initial material parameters, the neutronics calculation yields the effective neutron multiplication factor k_{eff} and the volumetric power distribution, which drives the FEniCS thermal–structural computation. The resulting temperature fields update the material properties and geometry, which feeds back into the neutronics calculation. Iteration continues until k_{eff} and the temperature field converge to a self-consistent steady state.

Physical coupling pathways.

The coupled HPR system exhibits two distinct physical pathways through which material-parameter uncertainty propagates to safety-critical outputs (Fig. 1):

1. **Stress pathway**: the SS316 monolith thermal conductivity ($k_{\mathrm{ref,SS316}}$) governs how efficiently heat is transported from fuel to heat pipe, setting local temperature gradients and the resulting thermal stress. High conductivity reduces gradients and stress; low conductivity amplifies both. This pathway has weak coupling to the neutronics feedback at first order.
2. **Reactivity pathway**: the thermal expansion coefficient (α_{base}) controls core geometric expansion under heating, which modifies neutron leakage and thereby k_{eff}. In the coupled steady state, materials with high α_{base} expand the core, reduce leakage, raise k_{eff}, and in turn increase local power density.

These pathways interact through a negative feedback loop in the coupled iteration: high thermal expansion raises k_{eff} and power deposition, which partially offsets the stress increase through altered temperature distributions. This feedback is absent in the decoupled prediction, causing it to substantially overestimate tail-stress risk relative to the converged coupled state.

We define two fidelity levels: *decoupled single-pass prediction*—the outputs of a single, uncoupled simulation pass without neutronics–thermal feedback; and *coupled steady-state*—the outputs of the fully converged iterative coupling. Each evaluation takes approximately 2000 s ($\sim$33 min) on a dedicated compute node.

A dataset of $n = 2900$ samples was generated by Latin hypercube sampling of the 8-dimensional input space, covering the full prior uncertainty range. Each sample provides 15 physics outputs: 7 decoupled quantities (average fuel temperature, maximum fuel temperature, maximum monolith temperature, maximum global stress, monolith temperature, core height after expansion, wall expansion) and 8 coupled quantities (k_{eff} plus the same 7 structural/thermal outputs). The dataset was split into training ($n_{\mathrm{train}} = 2029$), validation ($n_{\mathrm{val}} = 436$), and test ($n_{\mathrm{test}} = 435$) sets using a fixed random seed prior to any model selection.

4.3 Physics-regularized probabilistic surrogate

4.3.1 Network architecture

The surrogate is a HeteroMLP [8] that maps the 8-dimensional material parameter vector $\boldsymbol{\theta}$ to a predictive mean $\boldsymbol{\mu}(\boldsymbol{\theta})$ and log-variance $\log \boldsymbol{\sigma}^2(\boldsymbol{\theta})$ for each of the 15 physics outputs. At test time, the predictive distribution for output j is $p(y_j \mid \boldsymbol{\theta}) = \mathcal{N}(\mu_j(\boldsymbol{\theta}), \sigma_j^2(\boldsymbol{\theta}))$.

4.3.2 Training objectives

The *data-only baseline surrogate* is trained by minimising the heteroscedastic Gaussian negative log-likelihood (NLL), without any structural constraint on input–output

[PLACEHOLDER] HPR reactor cross-section showing monolith structure, fuel rods, heat-pipe locations, and SS316 cladding. Annotate the material regions whose properties constitute the 8 uncertain inputs in Table 6.

Fig. 6 Cross-sectional geometry of the heat-pipe-cooled reactor used in this study. The coupled thermo-mechanical model resolves the monolith, fuel rods, and heat pipes; material properties of the SS316 cladding are the primary source of parametric uncertainty (Table 6).

relationships:

$$\mathcal{L}_{\mathrm{NLL}} = \frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{15} \left[\frac{(y_{ij} - \mu_{ij})^2}{2\,\sigma_{ij}^2} + \frac{1}{2} \log \sigma_{ij}^2 \right], \tag{1}$$

where $(y_{ij}, \boldsymbol{\theta}_i)$ are training samples.

The *monotone-regularized surrogate* augments Eq. (1) with a monotonicity regularization term. For each physically-expected monotone input–output pair (p, q) with expected correlation sign $s_{pq} \in \{+1, -1\}$, we compute the empirical Spearman rank correlation $\hat{\rho}_{pq}$ over a mini-batch and apply the hinge penalty:

$$\mathcal{L}_{\mathrm{mono}} = \lambda \sum_{(p,q)} \max(0,\ \delta - s_{pq}\,\hat{\rho}_{pq}), \tag{2}$$

where λ is a regularization weight and δ is a margin hyperparameter. The total loss is $\mathcal{L} = \mathcal{L}_{\mathrm{NLL}} + \mathcal{L}_{\mathrm{mono}}$.

4.3.3 Hyperparameter optimisation

Hyperparameters for both surrogate variants (number of hidden layers, hidden width, learning rate, batch size, w_{mono}) were independently optimised using Optuna [9] with 40 trials per model, maximising validation NLL. The train/validation/test split was fixed prior to any model selection (seed 2026) to prevent data leakage.

4.4 Forward uncertainty propagation

20 000 samples are drawn from the joint prior distribution of the 8 input parameters. For each sample $\boldsymbol{\theta}_i$, the surrogate returns the predicted mean $\boldsymbol{\mu}(\boldsymbol{\theta}_i)$ for all 15 outputs. The distribution of these predicted means constitutes the epistemic (model-mean) forward uncertainty propagation. The primary quantity of interest is the second-iteration

[PLACEHOLDER] HeteroMLP architecture diagram. Input: 8-dim $\boldsymbol{\theta}$; shared hidden layers; twin output heads for $\boldsymbol{\mu}(\boldsymbol{\theta})$ and $\log \boldsymbol{\sigma}^2(\boldsymbol{\theta})$ (15 outputs each). Annotate the Spearman-rank monotonicity penalty path for the monotone-regularized variant.

Fig. 7 Architecture of the heteroscedastic MLP (HeteroMLP) surrogate. A shared backbone maps the 8-dimensional material parameter vector $\boldsymbol{\theta}$ to twin output heads that jointly predict the mean $\boldsymbol{\mu}(\boldsymbol{\theta})$ and log-variance $\log \boldsymbol{\sigma}^2(\boldsymbol{\theta})$ for all 15 physics outputs. The monotone-regularized variant (Section 2.1) adds a Spearman-rank penalty term (Eq. (2)) to enforce physics-consistent monotonicity constraints during training.

maximum global stress; the design threshold is $\tau = 131\,\text{MPa}$—the yield strength of the SS316 stainless-steel monolith at reactor operating temperature [2] (main text; 110 MPa and 120 MPa thresholds in Appendix C). The exceedance probability is estimated as

$$P_{\text{exceed}}(\tau) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[\mu_{\text{stress}}(\boldsymbol{\theta}_i) > \tau]\,. \tag{3}$$

4.5 Global sensitivity analysis

First-order (S_1) and total-order (S_T) Sobol indices [10, 11] are estimated using the Jansen estimator with independent random sampling ($N_S = 4096$ samples per matrix), repeated over 50 independent replications. The mean and 90% confidence interval of each index are reported as the mean and normal-approximation interval ($\bar{S} \pm 1.645\,\text{SE}$) across replications, following the variance-decomposition framework of [11]. The monotone-regularized surrogate mean prediction $\mu_{\text{stress}}(\boldsymbol{\theta})$ and $\mu_{k_{\text{eff}}}(\boldsymbol{\theta})$ serve as the function evaluators, enabling sensitivity indices to be obtained at a fraction of the cost of direct simulation [2].

4.6 Observation-driven posterior calibration

For the observation-driven calibration step, **18 benchmark cases** are drawn from the held-out test split (`experiments_0404/_shared/fixed_split/`), balanced across three stress categories (6 low, 6 near-threshold, 6 high; true stress $\approx$ 102–198 MPa). A separate set of **10 high-stress feasible-region cases** (true stress 133–194 MPa, all exceeding the 131 MPa design threshold) is used to test posterior updating in the high-risk regime. "Observations" for each case are the high-fidelity outputs at

the corresponding test-split design point corrupted by additive Gaussian noise with $\sigma_{\text{obs}} = 2\%$ of the output magnitude, simulating a realistic single-point measurement.

For each case, MCMC [12] with a Gaussian likelihood is used to update the prior over the 4-parameter calibration subset $(E_{\text{intercept}}, \alpha_{\text{base}}, \alpha_{\text{slope}}, \nu)$. These are the four parameters identified by the Sobol analysis (Section 2.3) as the dominant drivers of the coupled stress and k_{eff} responses plus ν, which couples the elastic and thermal-expansion pathways. The remaining four parameters (E_{slope}, $T_{\text{ref,SS316}}$, $k_{\text{ref,SS316}}$, $k_{\text{slope,SS316}}$) are fixed at their nominal prior-mean values, reflecting a partial-observation scenario in which SS316 thermal properties are treated as known from prior material characterisation; this design choice is an acknowledged limitation discussed in Section 3.

Each MCMC run generates 1200 effective posterior samples after burn-in; measured acceptance rates fall in the range 0.49–0.59 (mean ≈ 0.55) across all benchmark cases. Posterior predictions are validated against the existing HF dataset via nearest-neighbour lookup (rather than a true HF rerun at the posterior mean); this constitutes a proxy validation whose limitations are discussed in Section 3. The posterior safety fraction is defined as

$$P_{\text{safe}}(\tau) = P(\mu_{\text{stress}}(\boldsymbol{\theta}) < \tau \mid \text{data}) = 1 - P_{\text{exceed}}(\tau \mid \text{data}), \tag{4}$$

estimated from the posterior predictive samples.

5 Conclusion

This work presents a probabilistic surrogate framework for coupled multi-physics uncertainty quantification in a heat-pipe-cooled reactor, demonstrating four principal findings:

1. **Physics regularization improves generalization.** The `data-mono-ineq` HeteroMLP, trained with combined Spearman-rank monotonicity and physical-inequality constraints, matches or slightly improves in-distribution accuracy over the unregularized baseline (test NLL 0.319 vs. 0.407; stress $R^2 = 0.941$ vs. 0.938) while substantially improving OOD performance for α_{base} extrapolation (R^2: $0.774 \rightarrow 0.944$). This supports embedding domain knowledge as a soft regularizer.
2. **Multi-physics coupling compresses uncertainty.** Forward propagation shows that coupled steady-state outputs have significantly tighter distributions than decoupled predictions: an approximately 29% reduction in stress standard deviation and a more than two-orders-of-magnitude compression of k_{eff} spread. These are consequences of the neutronics–thermal feedback mechanism and cannot be captured by single-physics surrogates.
3. **Distinct sensitivity pathways exist.** Sobol analysis identifies $E_{\text{intercept}}$ as the dominant stress driver ($S_1 = 0.586$) and α_{base} as the dominant k_{eff} driver ($S_1 = 0.771$), indicating that design interventions for stress reduction and reactivity control can be treated with largely separable strategies at the dominant-factor level.

4. **Posterior calibration sharpens risk assessments.** Bayesian calibration on 18 benchmark test cases attains a mean 90% credible interval coverage of 0.875 over the four calibrated parameters; for 10 high-stress feasible-region cases (true stress 133–194 MPa) the posterior exceedance probability $P(\sigma > 131\,\text{MPa} \mid \text{obs})$ falls in the range 0.60–1.00, with 4 of 10 reaching certainty ($P \approx 1.00$), demonstrating that the pipeline correctly identifies high-risk parameter regions from observational evidence.

The computational speedup ($>10^7$ over the coupled HF solver) makes this framework practical for iterative design exploration and real-time risk monitoring in reactor systems with significant material uncertainty.

Acknowledgements. The authors gratefully acknowledge the support provided by the National Natural Science Foundation of China [12175138]; Shanghai Rising-Star Program; and LingChuang Research Project of China National Nuclear Corporation.

Declarations

- **Funding:** This work was supported by the National Natural Science Foundation of China [12175138]; Shanghai Rising-Star Program; and LingChuang Research Project of China National Nuclear Corporation.
- **Conflict of interest:** The authors declare no competing interests.
- **Data availability:** The dataset and trained model checkpoints will be made available upon reasonable request.
- **Code availability:** Analysis code will be made available upon reasonable request.
- **Author contributions:** [TBD]

Appendix A Out-of-distribution evaluation: both surrogates

Out-of-distribution performance was evaluated by constructing OOD sub-sets from the held-out test set: for each input feature in turn, samples with feature values in the upper or lower ∼20% of the training marginal were grouped as the OOD set; the remaining samples formed the in-distribution comparison. Table A1 reports test-set R^2 for coupled steady-state maximum global stress and k_{eff} under each OOD feature, comparing the unregularized baseline and the `data-mono` (Spearman-monotonicity-only) surrogate. Because `data-mono-ineq` (the main-text model) adds explicit physical-inequality constraints on top of the same monotonicity term, its OOD performance is expected to be at least as strong; the values below therefore serve as a conservative lower bound.

The largest improvement is for α_{base} extrapolation (stress $\Delta R^2 = +0.170$), consistent with the enforcement of the α_{base}–stress monotonicity constraint. For k_{eff} under α_{base} extrapolation, both models show substantial degradation ($R^2 \approx 0.29$), reflecting that extreme values of the dominant reactivity driver carry the model far outside the training regime; the monotonicity regularisation does not appreciably help in this case because the α_{base}–k_{eff} relationship is nonlinear over the extrapolation range. For ν, the

regularized model shows a slight stress R^2 decrease relative to baseline (-0.005); the monotonicity constraint for this pair is less tight, and the difference is within expected sampling variability.

Table A1 OOD evaluation: coupled steady-state maximum global stress R^2 and k_{eff} R^2 for the unregularized baseline and the `data-mono` surrogate, by OOD input feature. ΔR^2 = data-mono − baseline (stress).

OOD feature	n_{OOD}	Stress R^2		k_{eff} R^2	
		Baseline	data-mono	Baseline	data-mono
$E_{\text{intercept}}$	577	0.940	0.949	0.814	0.826
α_{base}	580	0.774	**0.944**	0.286	0.285
ν	575	0.922	0.917	0.590	0.623
α_{slope}	576	0.909	0.934	0.708	0.706

Source: `experiments_phys_levels/paper_ood_multi_feature_summary.csv`. OOD sets cover the upper/lower ∼20% tail of the training marginal for each feature. Values are from the `data-mono` surrogate (monotonicity regularisation only); the main-text `data-mono-ineq` model adds inequality constraints and is expected to perform at least comparably.

Appendix B Baseline surrogate: full test-set comparison

Table B2 compares the unregularized baseline and the `data-mono-ineq` surrogate on the held-out test set ($n = 435$) for the five primary outputs. In-distribution point accuracy is similar for the two models: coupled steady-state stress R^2 differs by only $+0.003$ (0.938 vs. 0.941) and RMSE by less than 0.2 MPa. The most notable improvements are in distributional quality—overall test NLL drops from 0.407 to 0.319 (a 22% reduction)—and in k_{eff} R^2 ($0.811 \rightarrow 0.830$), reflecting better multi-output joint calibration under physics regularisation. The primary motivation for selecting `data-mono-ineq` is therefore its improved calibration and substantially better OOD generalisation (Appendix A), not in-distribution stress accuracy.

Appendix C Predictive-draw risk variant and extended threshold sweep

As a supplementary uncertainty-decomposition perspective on the forward-UQ risk analysis of Section 2.2, we also compute a "predictive draw" variant in which each Monte Carlo sample is drawn from $y = \mu(\boldsymbol{\theta}) + \sigma(\boldsymbol{\theta})\,\varepsilon$ with $\varepsilon \sim \mathcal{N}(0, I)$, so that the learned aleatoric contribution is included in addition to the epistemic sampling over $\boldsymbol{\theta}$. Under the same nominal-design sampling ($N = 20\,000$, $\sigma_k = 1$) as in the main text, the coupled steady-state stress distribution has a mean of 162.6 MPa and a standard deviation of 32.15 MPa, compared with a mu-only standard deviation of approximately

Table B2 Comparison of the unregularized baseline and the `data-mono-ineq` (monotonicity + inequality regularized) surrogate on the held-out test set ($n = 435$), primary outputs only. Test NLL: baseline = 0.407, data-mono-ineq = 0.319.

Output		R^2		PICP_{90}	
	Pass	Baseline	d-m-ineq	Baseline	d-m-ineq
k_{eff}	Coupled	0.811	0.830	0.975	0.979
Max. fuel temp. (K)	Coupled	0.659	0.661	0.901	0.887
Max. monolith temp. (K)	Coupled	0.658	0.657	0.910	0.892
Max. global stress (MPa)	Coupled	0.938	0.941	0.936	0.947
Wall expansion (mm)	Coupled	0.988	0.998	0.982	1.000

R^2 from canonical training-time artifacts (`experiments_0404/models/{baseline,data-mono-ineq}/artifacts/metrics_*_fixed.json`); PICP_{90} from an independent re-evaluation pass (`fixed_eval/metrics_per_output_fixed.csv`). RMSE for stress: baseline = 7.58 MPa, data-mono-ineq = 7.38 MPa.

SI30.5MPa; including the aleatoric term therefore widens the spread by less than 2 MPa and does not overturn the risk ranking or the qualitative conclusions reported in the main text. Table C3 extends the threshold sweep to two higher design-relevant thresholds, $\tau = 169$ MPa (the MegaPower-class reported peak monolith stress under normal operating conditions [2]) and $\tau = 200$ MPa (a conservative tail-risk reference).

Table C3 Exceedance probability $P(\sigma > \tau)$ for the `data-mono-ineq` surrogate under the nominal-design prior at $\sigma_k = 1$, using either mu-only (epistemic, main text) or predictive-draw (including aleatoric) sampling.

Sampling	$\tau = 110$ MPa	$\tau = 120$ MPa	$\tau = 131$ MPa	$\tau = 169$ MPa	$\tau = 200$ MPa
mu-only (main text)	0.966	0.927	0.853	0.401	0.111
predictive draw	0.959	0.914	0.836	0.403	0.126

Source: `experiments_0404/risk_propagation_0410/data-mono-ineq/D1_muonly.json` (mu-only) and `D1_predictive_extended.json` (predictive draw); all values computed directly from 20 000 Monte Carlo samples at all five thresholds.

Practical interpretation.

Including the aleatoric term preserves the exceedance-probability ordering at all five thresholds and shifts each P_{exceed} by less than 0.02; qualitatively, the stress distribution remains concentrated above 131 MPa and the probability of exceeding the MegaPower reference of 169 MPa remains near 0.40 under the nominal prior. The main-text narrative—that the nominal design sits above the yield-threshold on average and benefits strongly from observation-driven posterior sharpening—is therefore robust to the choice of mu-only versus predictive-draw sampling.

Appendix D Full Sobol results

Table D4 provides first-order (S_1) Sobol indices with 90% CIs for all input–output pairs, comparing the unregularized baseline surrogate and the `data-mono-ineq` regularized surrogate used throughout the main text.

Table D4 First-order (S_1) Sobol indices for coupled steady-state maximum global stress and k_{eff}, comparing the unregularized baseline and the `data-mono-ineq` regularized surrogate. CIs are 90% bootstrap intervals; entries whose CI includes zero are marked with (†).

Output	Factor	S_1 (baseline)	S_1 (d-m-ineq)	CI low (d-m-ineq)	CI high (d-m-ineq)
Stress	E_{slope}	0.053	0.056	0.053	0.060
	$E_{\mathrm{intercept}}$	0.613	0.586	0.584	0.588
	ν	0.051	0.064	0.060	0.067
	α_{base}	0.157	0.169	0.165	0.172
	α_{slope}	0.024	0.033	0.030	0.036
	$T_{\mathrm{ref,SS316}}$	0.024	0.017	0.013	0.020
	$k_{\mathrm{ref,SS316}}$	0.065	0.053	0.050	0.056
	α_{SS316}	$0.001^{\dagger}$	$0.002^{\dagger}$	−0.001	0.005
k_{eff}	E_{slope}	$0.003^{\dagger}$	0.003	0.000	0.006
	$E_{\mathrm{intercept}}$	0.004	0.003	0.000	0.006
	ν	0.016	0.021	0.018	0.024
	α_{base}	0.809	0.771	0.770	0.772
	α_{slope}	0.162	0.197	0.195	0.200
	$T_{\mathrm{ref,SS316}}$	0.006	0.006	0.003	0.009
	$k_{\mathrm{ref,SS316}}$	0.009	0.010	0.007	0.013
	α_{SS316}	0.006	0.005	0.002	0.008

Source: `experiments_0404/sensitivity/{baseline,data-mono-ineq}/sobol_results.csv`. Baseline = unregularized heteroscedastic MLP; d-m-ineq = `data-mono-ineq` (Spearman monotonicity + physical-inequality regularization).

Appendix E Per-output accuracy table (all 15 outputs)

The full per-output accuracy metrics for the `data-mono-ineq` surrogate on the held-out test set are provided in Table E5; primary outputs used in the main-text risk, sensitivity and posterior analyses are marked ($\star$).

Appendix F Speedup benchmark details

The computational speedup of the Regularized surrogate versus the coupled HF solver is summarised in Table F6.

Table E5 Complete per-output accuracy metrics for the `data-mono-ineq` surrogate on the held-out test set ($n = 435$). Primary outputs marked ($\star$).

Output	Pass	RMSE	R^2
Avg. fuel temp. (K)	Decoupled	4.19	0.839
Max. fuel temp. (K)	Decoupled	7.15	0.843
Max. monolith temp. (K)	Decoupled	8.33	0.840
Max. global stress (MPa)	Decoupled	10.51	0.941
Monolith new temp. (K)	Decoupled	1.47	0.831
Core height after (mm)	Decoupled	0.057	0.893
Wall expansion (mm)	Decoupled	1.3×10^{-3}	0.997
k_{eff} ($\star$)	Coupled	2.9×10^{-4}	0.830
Avg. fuel temp. (K)	Coupled	2.27	0.661
Max. fuel temp. (K) ($\star$)	Coupled	4.23	0.661
Max. monolith temp. (K) ($\star$)	Coupled	4.79	0.657
Max. global stress (MPa) ($\star$)	Coupled	7.38	0.941
Monolith new temp. (K)	Coupled	0.64	0.662
Core height after (mm)	Coupled	0.600	0.073
Wall expansion (mm) ($\star$)	Coupled	1.1×10^{-3}	0.998

Source: `experiments_0404/models/data-mono-ineq/artifacts/metrics_data-mono-ineq_fixed.json`. MAE, PICP_{90} and MPIW_{90} for the five primary outputs are reported in the main-text Table 1; full distributional metrics for the secondary outputs are available in the evaluation manifest `fixed_eval/eval_manifest_fixed.json` (independent re-evaluation pass, values for stress/k_{eff} agree with the canonical training-time metrics to within ± 0.015 in R^2). The low coupled R^2 for *Core height after* reflects a narrow dynamic range in which the surrogate variance dominates, not a failure of the primary safety-metric prediction.

Table F6 Computational speedup of the surrogate versus the HF solver.

Metric	Value	Notes
Single-sample CPU latency	1.55×10^{-4} s	Regularized, CPU
Per-sample GPU batch latency	1.70×10^{-7} s	Batch size 1024
HF solver (one case)	$\sim$2000 s	OpenMC + FEniCS, 1 CPU node (user-confirmed; verify with run log)
Speedup (single CPU sample)	$\approx 1.3 \times 10^{7}$	HF = 2000 s
Speedup (GPU batch per sample)	$\approx 1.2 \times 10^{10}$	HF = 2000 s

Source: `paper_speedup_benchmark.json`.

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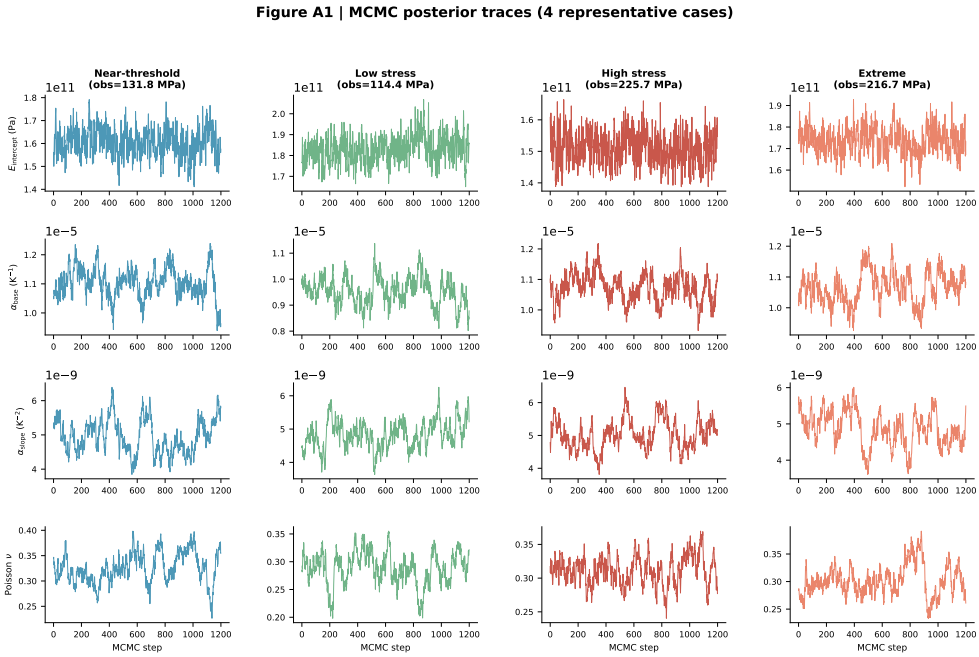

Fig. E1 MCMC posterior trace plots for four representative `data-mono-ineq` benchmark cases: low stress (observed ≈ 110 MPa), near-threshold (≈ 129 MPa), high stress (≈ 193 MPa), and a feasible-region probe case (≈ 185 MPa, drawn from the high-stress feasibility sweep). Rows correspond to the four calibrated parameters ($E_{\text{intercept}}$, α_{base}, α_{slope}, ν); columns to cases. Visual inspection confirms adequate chain mixing in all cases (acceptance rates 0.49–0.59).

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Figure A2 | Prior vs posterior marginal distributions

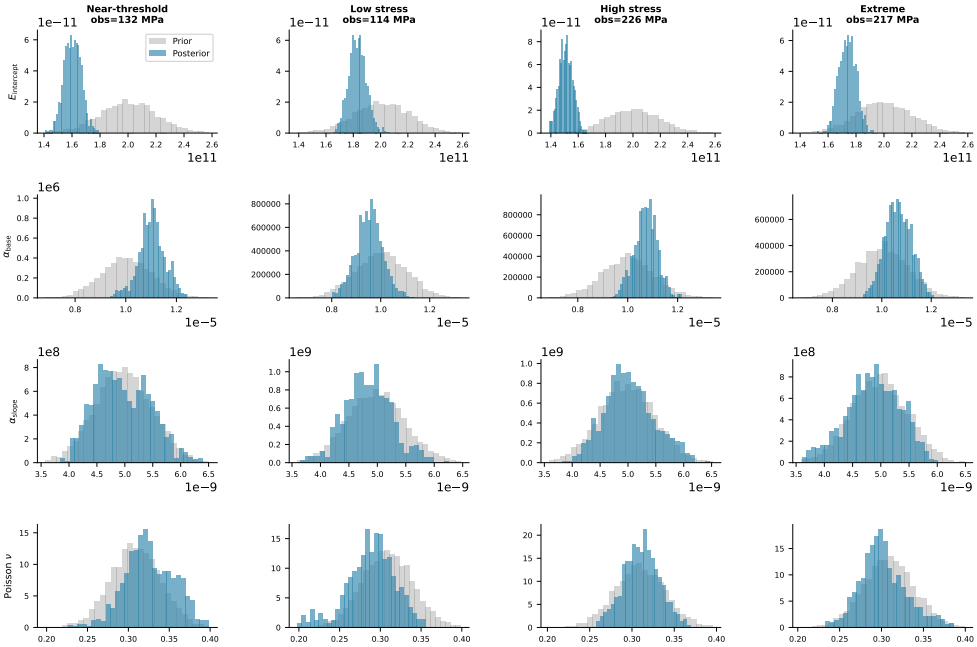

Fig. E2 Prior (grey) vs. posterior (blue) marginal distributions for the four calibrated parameters across four representative benchmark cases. Rows correspond to parameters; columns to cases (labelled with observed stress). Posterior contraction is most pronounced for α_{base} in the high-stress and extreme-stress cases, consistent with the Sobol analysis identifying α_{base} as the dominant reactivity driver.

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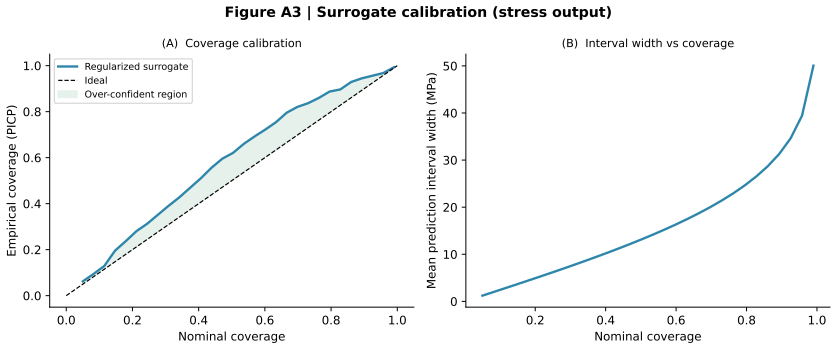

Fig. E3 Surrogate calibration diagnostics for the coupled steady-state maximum global stress output (`data-mono-ineq`, test set $n = 435$). (A) Reliability diagram: empirical prediction interval coverage probability (PICP) vs. nominal confidence level; the curve closely follows the ideal diagonal (dashed), confirming well-calibrated predictive uncertainty. (B) Mean prediction interval width as a function of nominal coverage.

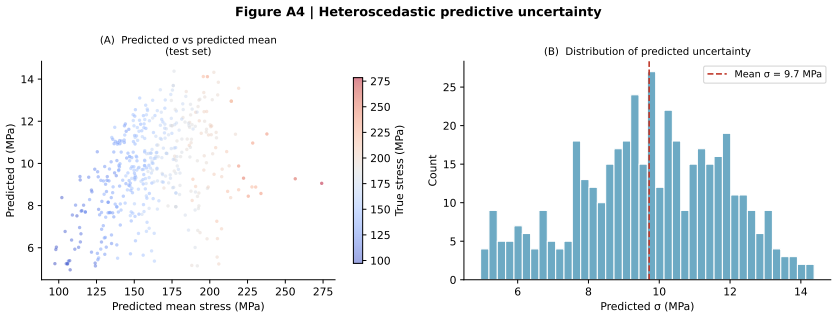

Fig. E4 Heteroscedastic predictive uncertainty for the coupled steady-state maximum global stress (`data-mono-ineq`, test set). (A) Predicted standard deviation σ as a function of predicted mean, coloured by true stress; σ increases in the high-stress regime, reflecting the model's recognition of higher aleatoric uncertainty in extreme parameter regions. (B) Distribution of predicted σ across test samples.

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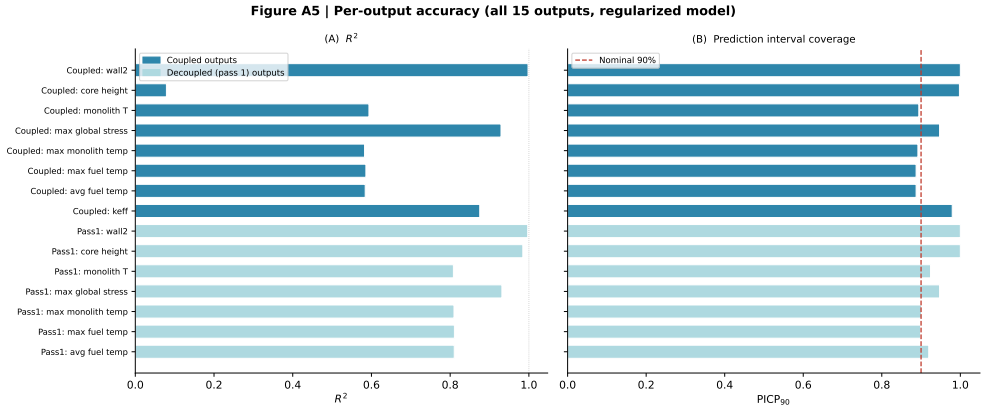

Fig. E5 Per-output accuracy for all 15 physics outputs (`data-mono-ineq` surrogate, test set). (A) R^2; (B) 90% prediction interval coverage ($PICP_{90}$). Blue bars: coupled steady-state outputs; light-blue bars: decoupled (decoupled single-pass) outputs. Wall expansion achieves $R^2 = 0.998$; coupled temperature outputs show lower $R^2 \approx 0.66$, reflecting their weaker sensitivity to the 8 material inputs in the coupled regime.

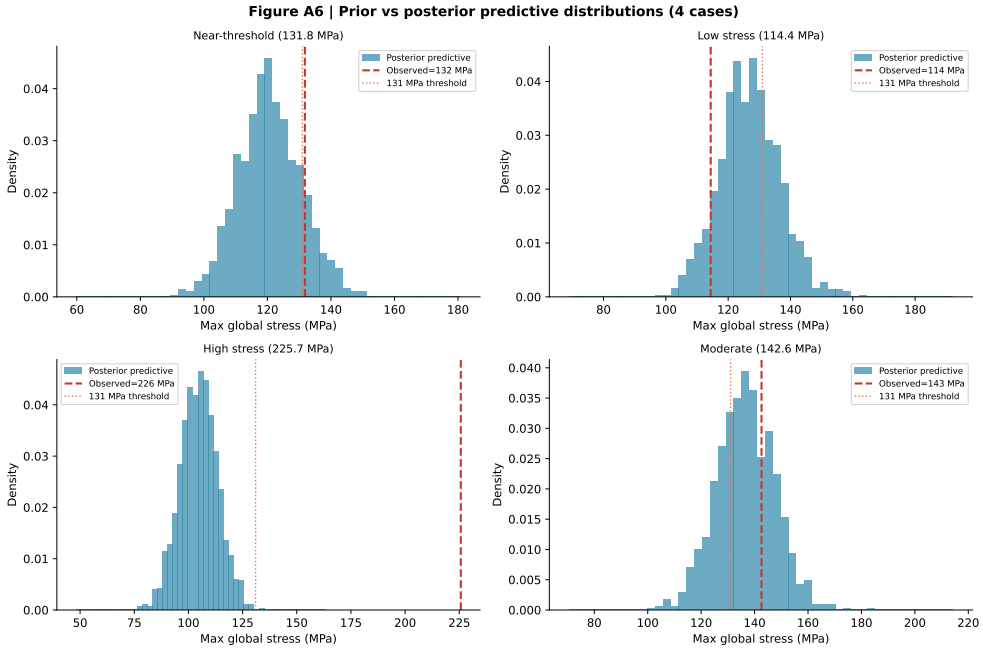

Fig. E6 Prior predictive (grey) vs. posterior predictive (blue) stress distributions for four representative benchmark cases (coupled steady-state max global stress). The observed stress (red dashed) and design threshold (orange dotted, 131 MPa) are indicated. Posterior updating concentrates probability mass near the observed value in all four cases, including the high-stress and extreme-stress scenarios.

Figure A7 | HeteroMLP architecture schematic

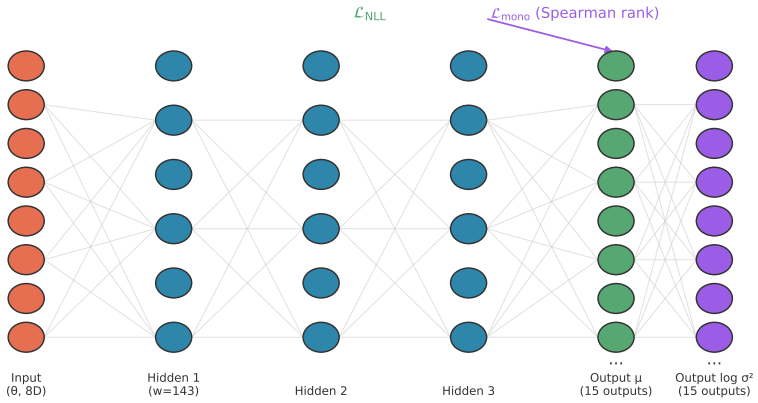

Fig. E7 Schematic of the heteroscedastic multilayer perceptron (HeteroMLP) architecture. The 8-dimensional material parameter vector $\boldsymbol{\theta}$ passes through shared hidden layers (width 143, depth 3 for Regularized) and is decoded into two heads: a mean output $\boldsymbol{\mu}$ and a log-variance output $\log \boldsymbol{\sigma}^2$, each of dimension 15. In the `data-mono-ineq` variant, both Spearman-rank monotonicity ($\mathcal{L}_{\text{mono}}$, Eq. 2) and physical-inequality constraints are applied during training.

Figure A8 | Training curves (NLL vs epoch)

□□□□□ Re-run run_train_fixed_surrogates.py to generate training_history_level{0,2}.csv (now supported by the updated script).

Fig. E8 Training curves: negative log-likelihood (NLL) vs. epoch for the baseline (Baseline) and physics-regularized (Regularized) surrogates. (*Placeholder shown; regenerate with actual training history files once available by re-running* `run_train_fixed_surrogates.py`.)